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Regiodivergent Reactions of 2-Alkynyl Aziridines with Grignard Reagents

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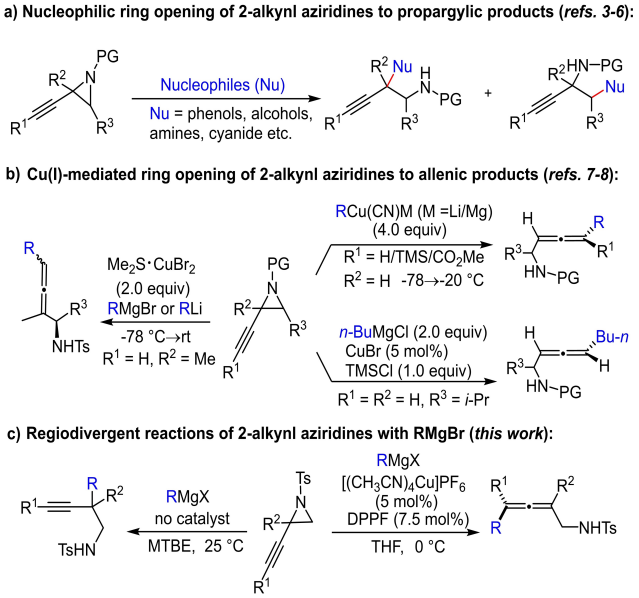
Abstract: Herein, we report regiodivergent nucleophilic attack of Grignard reagents to 2-alkynyl aziridines. Under the catalysis of a Cu(I)-diphosphine complex, benzylic Grignard reagents add to the carbon-carbon triple bond of the aziridines to deliver α-allenylamines in moderate to excellent yields; the direct addition of the Grignard reagents to the aziridines in the absence of catalyst provided homopropargylic amine derivatives. Both reactions could proceed in highly enantiospecific ways. Further synthetic manipulations of these products enable rapid access to diverse useful heterocyclic molecular scaffolds.

Keywords: aziridines; ring opening; Grignard reagents; alkynes; allenes

Introduction

Owing to the presence of both the strained aziridine ring and the alkyne bond, 2-alkynl aziridines represent a particular class of synthetic building blocks with multiple reactive sites for accessing diverse useful nitrogen-containing molecules while attracting much research intention.[1] Whilst the nucleophilic ring opening reactions of 2-alkyl or aryl aziridines usually proceed via an S_N2-type mechanism,[2] the reaction with 2-alkynyl aziridines may proceed via S_N2-type and/or S_N2'-type mechanisms to provide alkynyl products and/or allenic products. Oxygen nucleophiles,[3] especially phenols, have been reported to attack the aziridine ring to predominantly provide alkynyl products with elegant applications to the syntheses of complex bioactive molecules such as ustiloxin[4] and phomopsin B.[5] Relevant reaction systems with other nucleophiles such as nitrogen- and sulphur-nucleophiles have also been developed with varied degrees of success (Scheme 1a).[3c,6]

On the other hand, highly regioselective and enantiospecific nucleophilic attacks at the carbon-carbon triple bond in 2-alkynyl aziridines (mostly bearing a terminal alkyne) to generate allenic products have been realized with traditional reactive organometallic reagents



Scheme 1. Nucleophilic ring opening reactions of 2-alkynyl aziridines.

such as organolithium and organomaganesium under the mediation of stoichiometric amounts of Cu(I) salts

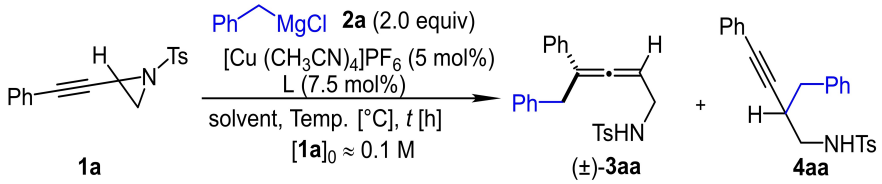
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(Scheme 1b), which typically require cryogenic conditions.[7–8] In contrast, relevant catalytic reaction systems remained underdeveloped while two isolated examples of Cu(I)-catalyzed ring opening reactions with *n*-BuMgCl in the presence of 1.0 equiv. of TMSCl have been reported.[7b] Notably, sporadic examples of Cu(I)-catalyzed borylative[9] and silylative[10] transformation of 2-alkynyl aziridines to allenic products have also been recently reported.[11] We report herein regiodivergent reactions of 2-alkynyl aziridines with Grignard reagents. With a catalytic amount of a Cu(I) salt, the reaction provided β-allenylamine products with good to excellent yields, whereas in the absence of the Cu(I) salt, homopropargylic amine products were obtained instead (Scheme 1c).

Results and Discussion

We initiated the study by screening the reaction conditions for the model reaction between aziridine **1a** and the Grignard reagent **2a** (Table 1). After a preliminary screening of metal salts (see Table S1 in the SI for details), $[Cu(CH_3CN)_4]PF_6$ was selected to evaluate different phosphine ligands (entries 1–6). In general, the presence of a phosphine ligand directed the reaction to the formation of the allenic product **3aa** with high selectivity and DPPF turned out to be the optimum ligand (Table 1, entry 5). Notably, previous relevant studies have been mainly restricted to aziridines bearing a terminal alkyne moiety,[7–8] and the current study first incorporates aziridines bearing an aryl alkyne into such a transformation. Unfortunately, the use of several chiral phosphine and BOX ligands hardly provided any chiral induction in both the product and the recovered aziridine (see Table S1 in the SI for details). An obvious solvent effect on the reaction was also observed: the use of toluene, MTBE and Et_2O as solvent all led to inferior results in terms of both yield and product selectivity (Table 1, entries 7–9). Decreasing the amount of either the ligand or the catalyst combination or increasing the reaction temperature led to inferior product selectivity (Table 1, entries 10–12). In the absence of a ligand, the S_N2-type ring opening product **4aa** was formed as major product, which could be obtained almost quanti-

Table 1. Screening of reaction conditions.[a]



Entry	L	Solvent	Temp. [°C]	*t* [h]	Yield [%][b]	
					3aa	**4aa**
1	PPh_3	THF	0	2	78	7[d]
2	PCy_3	THF	0	2	91	5[d]
3	*t*-BuXPhos	THF	0	2	87	5[d]
4	*rac*-BINAP	THF	0	2	81	4[d]
5	DPPF	THF	0	2	93	4[d]
6	$(p\text{-Tolyl})_3P$	THF	0	2	73	9[d]
7	DPPF	toluene	0	2	32	23
8	DPPF	MTBE	0	2	30	21
9	DPPF	Et_2O	0	2	32	20
10	DPPF (6 mol%)	THF	0	2	75	12
11[c]	DPPF	THF	0	6	70	15
12	DPPF	THF	25	1	45	17
13	DPPF	THF	-15	13	87	8[d]
14	–	THF	25	72	5[d]	71
15	–	toluene	25	72	11	89
16	–	MTBE	25	72	trace	98
17	–	Et_2O	25	96	14	82

[a] Reaction scale 0.1 mmol. The Grignard reagent was used as solution in THF.
[b] Isolated yields were reported unless noted otherwise.
[c] Run with 2.5 mol% of $[Cu(CH_3CN)_4]PF_6$ and 3.75 mol% of DPPF.
[d] Yields were determined by ^1H NMR analysis using *p*-bromobenzaldehyde as internal standard. Ts: *p*-toluenesulfonyl; BINAP: 1,1′-binaphthyl-2,2′-diphemyl phosphine; *t*-BuXPhos: 2-di-*tert*-butylphosphino-2′,4′,6′-triisopropylbiphenyl; DPPF: 1,1′-bis(diphenylphosphino)ferrocene; MTBE: methyl *t*-butyl ether.

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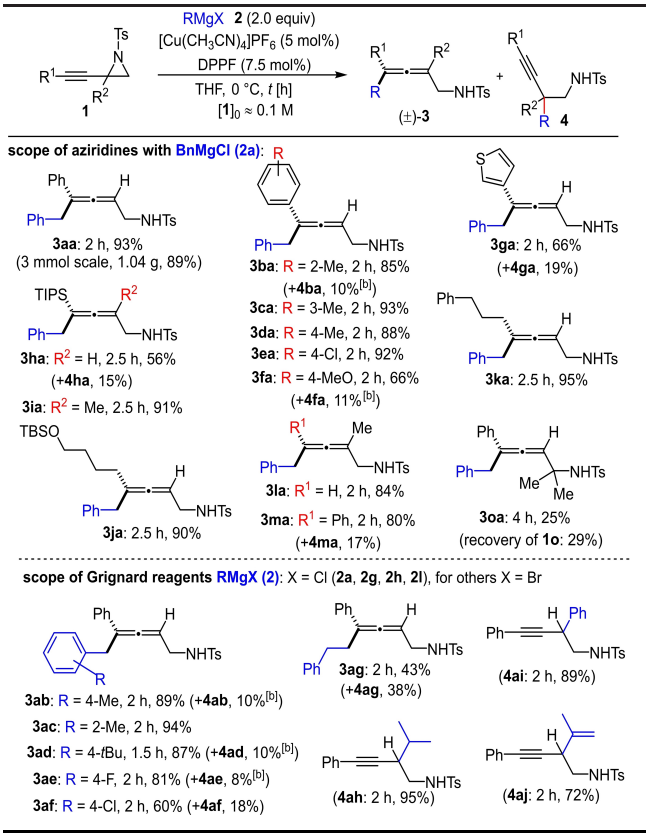
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tatively when the reaction was performed in MTBE at ambient temperature (Table 1, entry 16).

Subsequently, we examined the reaction scope with regard to the formation of the allenic products **3** (Table 2). In general, the reaction tolerated well a range of aziridines bearing (hetero)aryl-, silyl- and alkyl-substituted and terminal alkyne moieties to provide the desired products in high yields. In some cases, in addition to the allenic products, the S_N2-type ring-opening products were also obtained in considerable amounts (**4 fa**, **4 ga** and **4 ha**). The gram-scale preparation of the product **4 aa** in a comparable yield demonstrates the practicality of this reaction. Notably, 2,2-disubstituted propargylic aziridine disfavoured the S_N2-type ring opening while preferring the formation of the tetrasubstituted allenic product (**3 ha** *vs* **3 ia**). It's worth noting that the N-Boc counterpart of aziridine **1a** was not tolerated in this reaction in that only the aziridine (**1n**) was recovered (55%) after a prolonged reaction time of 48 h, together with the formation of unidentified complex mixture. The aziridine **1o** bearing two methyl groups at the 3-position also gave a poor yield of the desired product together with the recovery of starting material and the formation of unidentified mixtures. As to the scope of Grignard reagents in the reaction with aziridine **1 a**, benzylic Grignard reagents bearing different substituents on the benzene ring were suitable for the reaction to provide the allenic products in generally high yields. Surprisingly, the reaction with 4′-cholorobenzylmagnesium bromide gave 18% of homopropargylic amine product **4 af** together with 60% of **3 af**. For the primary alkyl Grignard reagents, the regioselectivity for the ring opening is poor in that the two types of products were obtained in comparable yields (**3 ag** and **4 ag**). In the case of isopropylmagnesium bromide, only the homopropargylic amine product **4 ah** was obtained in 95% yield. The use of corresponding phenyl or isopropenyl Grignard reagents all furnished the alkynyl products as the major products (**4ai** and **4aj**). Notably, alkyl Grignard reagents have been shown to undergo S_N2'-type reaction with propargylic ammonium salts with excellent regioselectivity.[12] The use of ethynylmagnesium bromide (**2k**) in this reaction with the aziridine **1 a** hardly produced any desired product while 57% of the aziridine was recovered together with unidentifiable complex mixture. The use of either 2-methylallylmagnesium chloride (**2l**) or the organolithium reagent *n*-BuLi produced a complex reaction mixture.[13] The structure of the product **3 ca** was determined by single crystal x-ray crystallographic analysis (CCDC 2391017).[14]

Next, the scope of the S_N2-type ring opening reaction of 2-alkynyl aziridines was investigated (Table 3). For aziridines bearing aryl alkyne moieties, the reaction proceeded smoothly to provide the desired homopropargylic amine products in good to excellent yields (**4 aa**–**4 ga**). The reaction could also be performed on gram-scale as demonstrated in the case of **4 aa**. While good yields could be obtained with aziridines bearing silyl alkynes, the reaction with those bearing alkyl-substituted alkynes provided allenic products (**3 ja** and **3 ka**) as major products. The aziridine **1 l** bearing a terminal alkyne was also not tolerated in this reaction to provide the allenic product (**3 la**) in a low yield together with unidentifiable complex mixture. The use of substituted benzylic Grignard reagents generally provided moderate yields in this reaction while the corresponding allenic products and/or other unidentified side products were formed concomitantly (**4 ab**–**4 af**). In addition, alkyl and aryl Grignard reagents provided low yields of the desired products (**4 ag**–**4 ai**) together with unidentifiable complex mixture. It's worth noting that in the presence of the Cu(I)-dppf complex, the homopropargylic amine products **4 ah** and **4 ai** were obtained in high yields (Table 2). Vinyl (**2j**) and ethynyl (**2k**) Grignard reagents were not tolerated in this reaction to give complex

Table 2. Reaction scope of the allenic products **3**.[a]



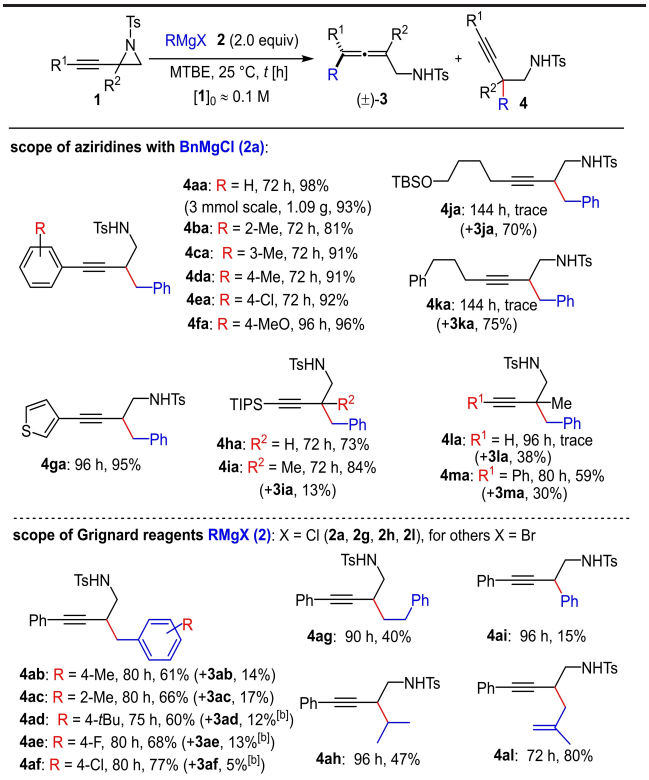
[a] Reaction scale 0.1 mmol. The Grignard reagents RMgX (X=Br/Cl) were used as solution in THF/Et_2O. Unless noted otherwise, isolated yields were reported and product ratios (**3**:**4**) are ≥ 20:1 as estimated by ^{1}H NMR analysis of the crude reaction mixture.

[b] Yields were determined by ^{1}H NMR analysis by using *p*-bromobenzaldehyde as internal standard.

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Table 3. Reaction scope of the alkynyl products **4**.[a]

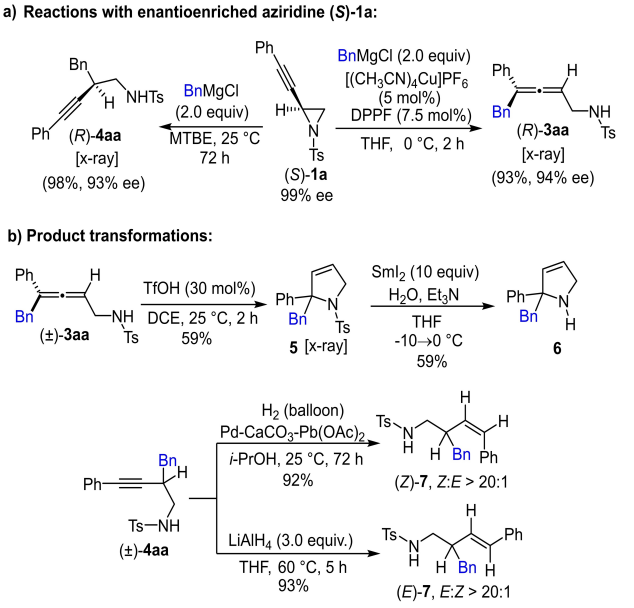


[a] Reaction scale 0.1 mmol. The Grignard reagents RMgX (X=Br/Cl) were used as solution in THF/Et_2O. Unless noted otherwise, isolated yields were reported and product ratios (**4**:**3**) are ≥20:1 as estimated by ^{1}H NMR analysis of the crude reaction mixture.

[b] Yields were determined by ^{1}H NMR analysis by using *p*-bromobenzaldehyde as internal standard.

reaction mixtures whereas the allylic Grignard reagent **2l** provided the desired product **4al** in 80% yield. Such a result is reminiscent of early findings that allylic Grignard reagents tend to undergo 1,2-addition to α,β-unsaturated carbonyl compounds, which could be accommodated by a six-membered-ring transition state.[15]

Under the standard reaction conditions, the regiodivergent reactions between enantioenriched aziridine (*S*)-**1a** and the Grignard reagent **2a** proceeded smoothly to provide the desired products with high enantiospecificity (Scheme 2a). The absolute configurations of the two products were unambiguously confirmed by single crystal x-ray crystallographic analysis,[14] which supports that the allenic product (*R*)-**3aa** (CCDC 2391015) might be formed via an *anti*-S_N2'-type mechanism[7b] and the homopropargylic amine product (*R*)-**4aa** (CCDC 2391016) might be formed via an S_N2-type mechanism. In previous relevant studies on copper-catalyzed reactions between Grignard reagents and propargylic electrophiles involving C–N bond cleavage,[12] the generation of Gilman-type organocuprates have been



Scheme 2. Experiments with enantioenriched (*S*)-**1a** and product transformations.

proposed and their tendency to attack carbon-carbon triple bond might be attributed to the "softer" nature of them according to hard and soft acid and base concept.[16] The α-allenylamine[17] product **3aa** could undergo an acid-catalyzed intramolecular hydroamination to produce compound **5**, the structure of which was confirmed by single crystal x-ray crystallographic analysis (CCDC 2391014).[14] Notably, when a sample of enantioenriched (*R*)-**3aa** (94% ee) was subjected to the otherwise same conditions, the product (*S*)-**5** was obtained in 59% yield and 85% ee. Detosylation of **5** with SmI_2 furnished the dihydropyrrole **6** in a moderate yield (Scheme 2b). Synthetically useful (*Z*)- and (*E*)-homoallylic amines **7** could be prepared in excellent yields from the homopropargylic amine product **4aa**.[18]

Conclusion

In summary, the regiodivergent ring opening reactions of 2-alkynyl aziridines with Grignard reagents has been investigated. Depending mainly on the use of a Cu(I) catalyst or the Grignard reagents, the ring opening reaction might proceed via either an *anti*-S_N2'-type pathway to provide α-allenylamine products or an S_N2-type mechanism to furnish homopropargylic amine products. The reaction is scalable and occurs in a highly enantiospecific fashion with enantioenriched 2-alkynyl aziridines.

Experimental Section

Typical procedure for the preparation of allenic products **3**: Under argon atmosphere, to an oven-dried Schlenk tube was

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added $[(CH_3CN)_4Cu]PF_6$ (1.9 mg, 0.005 mmol) and DPPF (4.2 mg, 0.0075 mmol). The system was evacuated under vacuum and back-filled with argon (repeated twice). Then 0.5 mL of anhydrous THF was added and the resulting mixture was stirred at rt for 30 minutes. The tube was then placed in a cold bath (0 °C) followed by the sequential addition of aziridine **1** (0.1 mmol), Grignard reagent (as solution in THF/Et_2O, 0.2 mmol) and 0.3 mL of anhydrous THF. The reaction mixture was stirred at 0 °C until the disappearance of **1** as monitored by TLC (visualized by staining with iodine and/or irradiation with UV light). Then 10 mL of saturated aqueous $NaHCO_3$ solution and 10 mL of ethyl acetate (EtOAc) were added, and the phases were separated. The aqueous phase was extracted with EtOAc (3×8 mL). The combined organic phases were dried over anhydrous Na_2SO_4 and concentrated via rotary evaporation under reduced pressure. Then the crude mixture was purified by column chromatography on silica gel (eluting with petroleum ether/EtOAc (v/v): 6:1–5:1) to furnish the desired products **3**.

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